

TORONTO METROPOLITAN UNIVERSITY

Master of Science in Data Science and Analytics

CRIME ARREST PREDICTION USING MULTI-SOURCE SPATIOTEMPORAL DATA AND MACHINE LEARNING TECHNIQUES

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Abstract

Crime prediction and arrest outcome prediction are important tasks for supporting law enforcement agencies in crime prevention and resource allocation. Traditional crime analysis often focuses on historical crime records alone and may overlook the influence of external factors such as socio-economic conditions, weather patterns, and human mobility. This research proposes a multi-source data-driven framework for predicting arrest outcomes using crime, socio-economic, weather, and public transit mobility data collected from the City of Chicago.

The study integrates the Chicago Crimes dataset with community-level socio-economic indicators, daily weather observations, and Chicago Transit Authority (CTA) ridership data. Several data preprocessing techniques were applied, including missing values handling, spatial data integration, feature engineering, and feature selection. The resulting dataset contains spatial, temporal, environmental, and mobility-related variables that provide a comprehensive representation of crime-related conditions.

To address the arrest prediction problem, multiple machine learning and deep learning models are evaluated, including Logistic Regression, Decision Tree, Random Forest, Extra Trees, Gradient Boosting, AdaBoost, XGBoost, LightGBM, CatBoost, Multi-Layer Perceptron (MLP), Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and Temporal Convolutional Network (TCN). Model performance is assessed using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and PR-AUC metrics. In addition, exploratory data analysis is conducted to investigate the relationship between crime patterns, socio-economic conditions, weather variables, and public transit mobility across different time periods, including the COVID-19 pandemic.

The findings of this study are expected to provide insights into the factors associated with arrest outcomes and demonstrate the effectiveness of integrating multi-source spatiotemporal data for crime-related predictive analytics. The proposed framework contributes to the development of more informed and data-driven approaches for public safety decision-making.

1. Literature Review

This section presents a comprehensive review of relevant scholarly papers and technical articles related to crime prediction, spatiotemporal modeling, arrest prediction, and multi-source data integration. Each paper is critically analyzed to identify contributions, limitations, and research gaps that motivate the proposed study.

Paper 1: Fan et al. (2025) – Research on a Crime Spatiotemporal Prediction Method Integrating Informer and ST-GCN [1]

Fan et al. (2025) proposed a hybrid crime prediction framework combining Informer and Spatial-Temporal Graph Convolutional Networks (ST-GCN) to predict crime occurrences in Chicago communities. The model utilized approximately 320,000 crime records from 22 police districts between 2015 and 2020 along with weather and holiday information. ST-GCN captured spatial and temporal dependencies. The proposed approach outperformed traditional machine learning and deep learning models in predicting assault, robbery, theft, and criminal damage crimes.

The study successfully integrated spatial and temporal dependencies using Informer and ST-GCN and achieved strong crime forecasting performance. However, it focused only on crime occurrence prediction and did not consider arrest outcomes. In addition, socio-economic and mobility factors were not included. This research addresses these limitations by incorporating multiple external data sources and focusing on arrest prediction.

Paper 2: Sankul et al. (2025) – Crime Data Analysis and Prediction of Arrest Using Machine Learning [2]

Sankul et al. (2025) explored the use of machine learning techniques for crime analysis and arrest prediction. The study employed historical crime records and considered factors such as crime type, location, time, socio-economic information, and environmental conditions. Random Forest was identified as the primary prediction model, while other machine learning algorithms were also discussed. The research demonstrated the usefulness of machine learning for assisting law enforcement agencies in decision-making and resource allocation.

This study is one of the few works that examined arrest prediction directly. However, it relied primarily on traditional machine learning models and lacked advanced spatiotemporal deep learning approaches. The proposed research extends this work by comparing both machine learning and deep learning models using a richer multi-source dataset.

Paper 3: Han et al. (2020) – Risk Prediction of Theft Crimes in Urban Communities: An Integrated Model of LSTM and ST-GCN [3]

Han et al. (2020) proposed an integrated crime prediction model combining Long Short-Term Memory (LSTM) networks and Spatial-Temporal Graph Convolution Networks (ST-GCN). The model was developed using Chicago crime data and incorporated weather and holiday information. LSTM learned temporal crime patterns. Experimental results demonstrated improved prediction performance for theft crime compared with existing approaches. The study highlighted the importance of combining spatial and temporal information for crime forecasting.

The integration of LSTM and ST-GCN effectively captured temporal and spatial crime patterns. However, the study focused only on theft crimes and did not investigate arrest outcomes. The proposed research considers all crime categories and evaluates arrest prediction using broader contextual information.

Paper 4: Riyazuddin et al. (2026) – Spatiotemporal Crime Prediction and Forecasting Using Machine Learning and Deep Learning Models [4]

Riyazuddin et al. (2026) introduced a Hierarchical Spatiotemporal Transformer (HSTT) for urban crime forecasting. The framework integrated structured crime data, mobility information, and external contextual factors to model crime patterns. The authors compared traditional machine learning methods with advanced deep learning models, including graph neural networks and transformer architectures. Results showed that HSTT achieved superior prediction performance while also addressing fairness, interpretability, and uncertainty estimation.

The HSTT framework demonstrated the effectiveness of transformer-based architectures for crime forecasting. Despite its strong predictive performance, the study focused on crime risk prediction rather than arrest outcomes. This research extends the concept by examining factors associated with arrests using integrated crime, socio-economic, weather, and mobility data.

Paper 5: – Crime Analysis Through Machine Learning [5]

This study focused on predicting crime types in Vancouver, Canada using machine learning techniques. The authors utilized crime records from the Vancouver Police Department and City of Vancouver Open Data Portal covering the period from 2003 to 2018. Features included temporal variables, neighborhood information, geographic coordinates, and crime history. K-Nearest Neighbor (KNN) and AdaBoost were used for multi-class crime classification. The study demonstrated the feasibility of predicting crime categories based on spatial and temporal characteristics. However, the model achieved relatively low prediction accuracy and did not incorporate socio-economic, weather, or mobility-related factors.

The study demonstrated that machine learning can be used to classify crime types using spatial and temporal variables. However, prediction accuracy remained relatively low and no external variables were considered. The proposed research incorporates additional contextual information to improve predictive performance.

Paper 6: – Exploring Spatio-Temporal and Cross-Type Correlations for Crime Prediction [6]

This paper proposed the CCC (Cross-Type and Spatio-Temporal Correlations) framework for crime prediction. Using New York City crime data, the study investigated relationships among different crime types, spatial regions, and temporal patterns. The model incorporated crime records, weather information, mobility data, POI data, public service complaints, and police activity records. Results showed that crime prediction accuracy improved significantly when cross-type, spatial, and temporal relationships were modeled together. The proposed CCC framework outperformed several baseline approaches.

The CCC framework effectively modeled spatial, temporal, and cross-type crime relationships. However, socio-economic variables and arrest outcomes were not considered. The current study addresses this limitation by integrating multiple external datasets and focusing on arrest prediction.

Paper 7: – Deep Learning for Real-Time Crime Forecasting and Its Ternarization (2019) [7]

This study developed a real-time crime forecasting framework using deep learning techniques. The authors adapted the Spatial-Temporal Residual Network (ST-ResNet) model to predict hourly crime intensity at the neighborhood level in Los Angeles. Historical crime records, weather

information, holidays, and temporal trends were used as inputs. The framework incorporated temporal and spatial enhancement techniques to improve prediction performance. Experimental results showed that the proposed model outperformed traditional forecasting approaches.

The study demonstrated the effectiveness of deep learning for real-time crime forecasting. However, it relied on a short observation period (six months) and ignored socio-economic and mobility variables. This research utilizes a longer historical dataset and incorporates additional explanatory factors.

Paper 8: – Prediction of Crime Occurrence from Multi-Modal Data Using Deep Learning (2017) [8]

This study proposed a deep learning framework for crime occurrence prediction using multiple data sources. The authors combined Chicago crime data with socio-economic variables, weather information, and street-view image features extracted using AlexNet. A DNN-based feature fusion network was developed to integrate spatial, temporal, environmental, and historical crime information. Results demonstrated that the proposed deep learning model significantly outperformed traditional machine learning approaches.

The use of multi-modal data significantly improved crime occurrence prediction. However, mobility information was not included and the focus remained on crime occurrence rather than arrests. The proposed research extends multimodal integration by incorporating public transit mobility data and arrest outcomes.

Paper 9: – Predictability Comparison of Three Kinds of Robbery Crime Events Using LSTM (2019) [9]

This study investigated the predictability of three robbery crime types—commercial, pedestrian, and residential robberies—using Long Short-Term Memory (LSTM) networks. Crime data from Atlanta and Baltimore were analyzed to examine how prediction performance changes across different spatial and temporal scales. The results showed that commercial and residential robberies were generally more predictable than pedestrian robberies. The study demonstrated the ability of LSTM to capture temporal crime patterns and long-term dependencies.

The findings showed that different robbery types exhibit different levels of predictability. However, the study focused exclusively on robbery crimes and omitted important contextual variables. This research examines all crime categories and integrates multiple external factors.

Paper 10: – Crime Spatiotemporal Prediction with Fused Objective Function in Time Delay Neural Network (2020) [10]

This study proposed an Enhanced Nonlinear Autoregressive with Exogenous Input (eNARX) neural network for serial crime prediction. The model aimed to predict future crime locations, crime timing, suspect age, and suspect nationality using historical crime records. The proposed eNARX model incorporated a new activation function (SiRBF) to improve learning performance. Experimental results showed that eNARX outperformed traditional BPTT and NARX models in prediction accuracy.

The eNARX model improved prediction accuracy for serial crime forecasting. However, its focus on serial offenders and limited dataset restricts its broader applicability. The proposed research investigates city-wide arrest prediction using a large-scale dataset.

Paper 11: – Deep Spatio-Temporal Residual Networks for Citywide Crowd Flows Prediction (ST-ResNet) [11]

This paper introduced the Spatio-Temporal Residual Network (ST-ResNet) to predict citywide crowd flow patterns. The model integrated spatial dependencies, temporal trends, weather conditions, and calendar information through a deep residual learning framework. Using taxi and bike mobility datasets from Beijing and New York City, the study demonstrated that ST-ResNet significantly outperformed traditional forecasting approaches. The research highlighted the importance of combining recent observations, daily periodic patterns, and long-term trends.

Although the study focused on crowd flow prediction rather than crime prediction, it introduced important spatiotemporal modeling concepts that have influenced crime forecasting research. The proposed study applies similar spatiotemporal principles to crime arrest prediction.

Paper 12: – Machine Learning in Crime [12]

This paper presented a systematic literature review of 68 crime prediction studies published between 2010 and 2022. The review examined datasets, machine learning algorithms, feature types, evaluation metrics, challenges, and future research directions in crime prediction. The

authors found that artificial neural networks, decision trees, random forests, and deep learning models were among the most commonly used techniques. The study highlighted key challenges such as data quality issues, class imbalance, limited public datasets, and lack of standardized evaluation methods.

The systematic review highlighted the growing importance of deep learning, spatiotemporal modeling, and multi-source data integration. The identified research gaps directly support the need for the proposed framework that combines multiple datasets and advanced predictive models.

Paper 13: – Examining Deep Learning Architectures for Crime Classification and Prediction [13]

This study investigated whether deep learning models can predict future crime hotspots more effectively than traditional machine learning techniques. Using crime datasets from Philadelphia, Seattle, Minneapolis, Washington DC, and San Francisco, the authors evaluated three deep learning architectures that combined CNN and LSTM networks. The proposed Spatial First Then Temporal (SFTT) model achieved the best performance by first learning spatial crime patterns and then temporal trends. Results showed that deep learning models consistently outperformed traditional machine learning methods.

The study demonstrated that deep learning models outperform traditional machine learning methods for hotspot prediction. However, it relied only on crime records and ignored external contextual factors. The proposed research incorporates socio-economic, weather, and mobility information to enhance prediction.

Paper 14: – Impact of COVID-19 on Crime: A Spatial Temporal Analysis in Chicago (2021) [14]

This study examined how the COVID-19 pandemic affected crime patterns across Chicago. Using crime data from 2016 to 2020, the authors analyzed changes in crime trends before, during, and after lockdown periods through temporal and spatial statistical methods. Results showed substantial decreases in theft, fraud, assault, battery, and robbery during lockdown periods, while burglary slightly increased. The study also identified significant geographic shifts in crime hotspots, particularly in downtown Chicago.

The study provided valuable insights into how COVID-19 changed crime patterns in Chicago. However, it was descriptive rather than predictive. This research extends those findings by incorporating mobility analysis and predictive modeling within the COVID-19 context.

Paper 15: Cohen and Felson (1979) – Social Change and Crime Rate Trends: A Routine Activity Approach [15]

This landmark study introduced Routine Activity Theory (RAT), one of the most influential theories in criminology. The authors argued that crime occurs when a motivated offender, a suitable target, and the absence of capable guardians converge in space and time. Using national crime statistics and social indicators, the study explained rising crime rates despite improvements in economic and social conditions. Rather than focusing on offender motivation, the theory emphasized crime opportunities created through everyday human activities.

Routine Activity Theory provides a strong theoretical foundation for understanding crime opportunities. However, it is a conceptual framework rather than a predictive model. The proposed research operationalizes these concepts using real-world mobility and environmental data.

Paper 16: – Crime Hotspot Forecasting: A Recurrent Model with Spatial and Temporal Information (2017) [16]

This study proposed a Spatio-Temporal Neural Network (STNN) for forecasting future crime hotspots. Using crime data from Portland, Oregon, the model combined spatial neighborhood information with historical temporal crime patterns to improve hotspot prediction. The proposed STNN architecture was compared with several machine learning and deep learning methods, including Random Forest, Logistic Regression, RNN, LSTM, and GRU models. Results demonstrated that incorporating both spatial and temporal information improved prediction performance.

The STNN framework demonstrated the importance of combining spatial and temporal information for hotspot prediction. However, the study excluded socio-economic, weather, mobility, and arrest-related variables. These limitations are addressed in the current research.

Paper 17: Zhou et al. (2021) – Informer: Beyond Efficient Transformer for Long Sequence Time-Series Forecasting [17]

This paper proposed Informer, an efficient Transformer-based architecture designed for long sequence time-series forecasting. The authors addressed limitations of traditional LSTM and Transformer models, particularly high computational complexity and memory requirements for long forecasting horizons. Informer introduced ProbSparse Self-Attention, Self-Attention Distilling, and a Generative Decoder to improve efficiency while maintaining forecasting accuracy. Experimental results demonstrated that Informer outperformed LSTM, ARIMA, Prophet, and several Transformer variants on multiple benchmark datasets.

Informer demonstrated exceptional performance for long-sequence forecasting and has influenced recent crime prediction frameworks. However, the study was not developed specifically for crime data. This research applies similar long-term temporal modeling concepts within a crime prediction context.

Paper 18: Wang et al. (2016) – Crime Rate Inference With Big Data [18]

This study investigated whether crime rates in Chicago could be estimated more accurately by integrating traditional demographic variables with urban big data sources such as Point-of-Interest (POI) information and taxi mobility data. The authors used Chicago crime records, census data, Foursquare POI data, and taxi trip information at the community area level. Results showed that both POI and mobility features significantly improved crime rate estimation compared to demographic variables alone. The study highlighted the importance of human mobility and urban activity patterns in understanding crime distribution.

The study provided strong evidence that mobility and urban activity patterns improve crime rate estimation. However, it relied on traditional regression models and did not consider arrest prediction. The proposed research builds upon these findings by integrating mobility data within advanced machine learning and deep learning frameworks for arrest prediction.

2. Data Description and Exploratory Data Analysis

2.1 Dataset Description

This study utilizes multiple datasets from different sources to develop a comprehensive framework for crime arrest prediction in the City of Chicago. The datasets include crime records, socio-economic indicators, weather information, mobility data, community area boundaries, and community reference information. The integration of these datasets enables the analysis of spatial, temporal, environmental, social, and mobility-related factors that may influence arrest outcomes.

2.1.1 Chicago Crime Dataset

The primary dataset used in this research is the Chicago Crime Dataset obtained from the Chicago Data Portal. The dataset contains reported crime incidents from 2001 to 2023 and includes detailed information regarding crime characteristics, location, arrest status, and temporal attributes. The crime dataset serves as the foundation of this study and provides the target variable (Arrest) used for predictive modeling.

Key variables include:

- Primary Type
- Description
- Arrest (target variable)
- Domestic
- Beat
- District
- Community Area
- FBI Code
- Latitude / Longitude
- Date and Time

2.1.2 Socio-Economic Dataset

To capture community-level social and economic conditions, socio-economic indicators were collected from publicly available Chicago community datasets. These indicators provide

information regarding economic status, education, and employment conditions across Chicago communities. The socio-economic data was integrated with crime records using community area identifiers and corresponding time periods.

Variables include:

- Per Capita Income
- Unemployment Rate
- Percentage Without High School Education

2.1.3 Weather Dataset

Weather data was obtained from NOAA daily weather records. Environmental conditions can influence both human behavior and crime opportunities; therefore, weather variables were included in the analysis. Additional weather indicators were created during preprocessing.

Variables include:

- Maximum Temperature (TMAX)
- Minimum Temperature (TMIN)
- Precipitation (PRCP)
- Snowfall (SNOW)
- Wind Speed (AWND)
- Average Temperature

Derived indicators:

- Rain Indicator
- Snow Indicator
- High Wind Indicator
- Temperature Category

2.1.4 Mobility Dataset

Human mobility information was obtained from Chicago Transit Authority (CTA) ridership data. Mobility variables provide insights into population movement patterns and urban activity levels. The mobility dataset was merged with crime records using daily timestamps.

Variables include:

- Bus Ridership
- Rail Boardings
- Total Daily Rides
- Day Type

2.1.5 Chicago Community Area Boundary Dataset

The Chicago Community Area Boundary dataset contains geographic boundary polygons representing the 77 official community areas of Chicago. The dataset was used to perform spatial analysis and spatial joins to identify community areas using latitude and longitude coordinates, recover missing community area values, support integration of socio-economic indicators, and enable spatial crime analysis.

Key variables include:

- Community Area Number
- Community Area Name
- Geographic Polygon Boundaries

2.2 Data Preprocessing

Several preprocessing steps were performed to improve data quality, handle missing values, and integrate multiple datasets into a unified analytical framework.

2.2.1 Missing Value Handling

Community Area Imputation: Community Area contained a substantial number of missing values. A spatial join was performed between crime incident coordinates and Chicago community area boundary polygons to assign community area values to records with valid geographic coordinates.

District Imputation: A small number of missing District values were identified. Since Beat and District are strongly related, missing District values were imputed using the most frequent District associated with each Beat.

Latitude and Longitude Imputation: Missing geographic coordinates were estimated using the average latitude and longitude values associated with each police Beat.

2.2.2 Spatial Data Processing

Spatial processing techniques were applied to improve data quality and integrate geographic information. Latitude and longitude coordinates were matched to community area boundary polygons using geographic information system (GIS) operations. This procedure enabled the recovery of missing community area values and supported the integration of socio-economic information.

2.2.3 Data Integration

The datasets were merged using common identifiers as follows:

- Crime and Weather data were merged using Date.
- Crime and Mobility data were merged using Date.
- Crime and Socio-Economic data were merged using Community Area and Year.
- Community names were merged using Community Area identifiers.

2.2.4 Feature Engineering

Several additional features were created from existing variables to enhance predictive modeling performance.

Temporal features:

- Year
- Month
- Day
- Day of Week
- Hour

2.2.5 Descriptive Statistics

Metric	Value
Total Records	7,970,658

Metric	Value
Total Features	34
Missing Values	0
Arrest Rate	25.8%
Non-Arrest Rate	74.2%
Study Period	2001–2023

2.3 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the characteristics of the integrated crime dataset and identify relationships among crime incidents, socio-economic conditions, weather factors, and mobility patterns. The analysis provides insights into temporal, spatial, environmental, and social factors that may influence crime occurrence and arrest outcomes.

2.3.1 Annual Crime Trends

Figure 1 presents the annual crime counts and arrest rates in Chicago from 2001 to 2023. The results indicate that crime incidents generally declined over the study period. While fluctuations occurred across different years, the overall trend suggests a reduction in reported crime compared to the early 2000s. The arrest rate remained relatively stable, although noticeable variations were observed during specific periods.

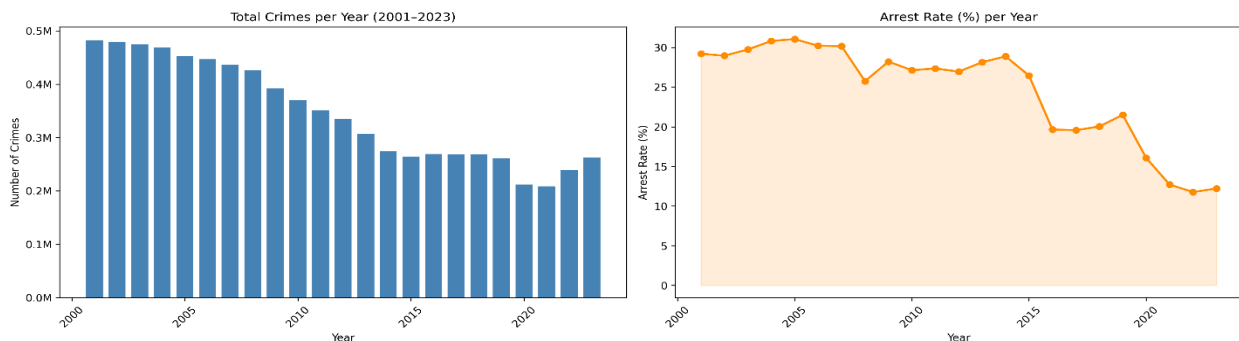


Figure 1: Crimes and Arrest Rate per Year

2.3.2 Monthly Crime Patterns

Figure 2 illustrates crime counts and arrest rates across different months of the year. The monthly analysis reveals seasonal variations in criminal activity. Certain months experienced higher crime volumes, potentially due to increased outdoor activity and favorable weather conditions. The arrest rate remained relatively consistent throughout the year, indicating stable law enforcement effectiveness across seasons.

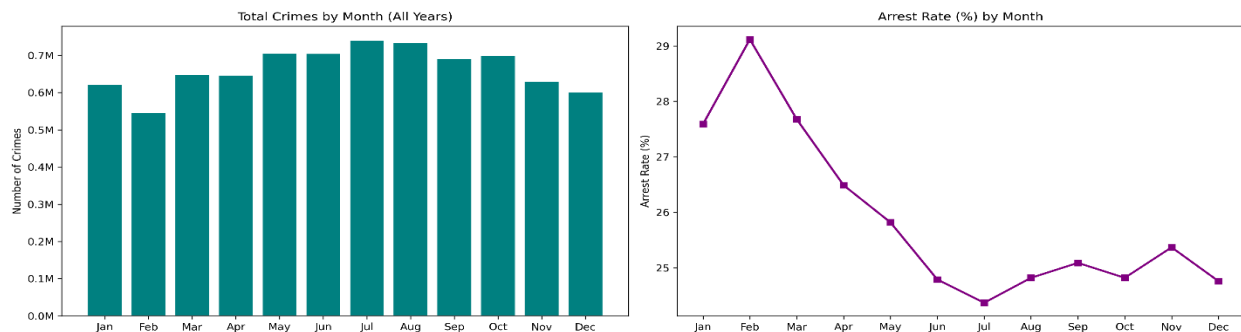


Figure 2: Crimes and Arrest Rate by Month

2.3.3 Crime Type Distribution

Figure 3 shows the most frequently reported crime categories. The analysis indicates that a small number of crime categories account for a large proportion of total incidents. Theft, battery, criminal damage, narcotics-related offenses, and assault represent some of the most frequently reported crimes.

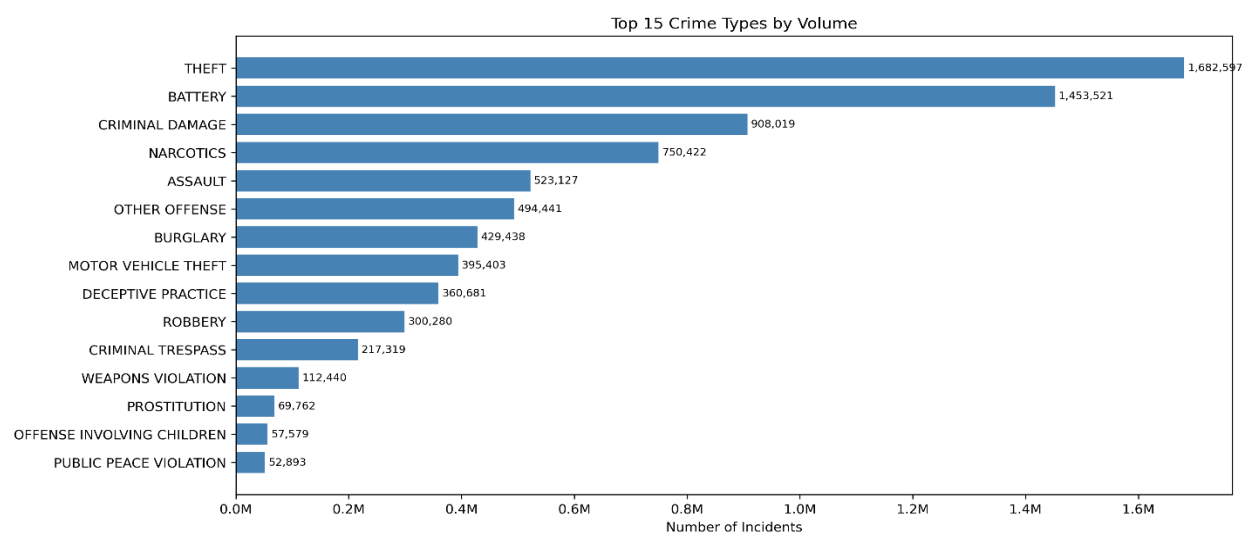


Figure 3: Top 15 Crime Types by Volume

2.3.4 Arrest Rate by Crime Type

Figure 4 compares arrest rates across major crime categories. Significant differences in arrest rates were observed among crime categories. Certain offenses demonstrated higher arrest rates due to stronger evidence availability or law enforcement priorities, whereas other crime categories exhibited lower arrest rates.

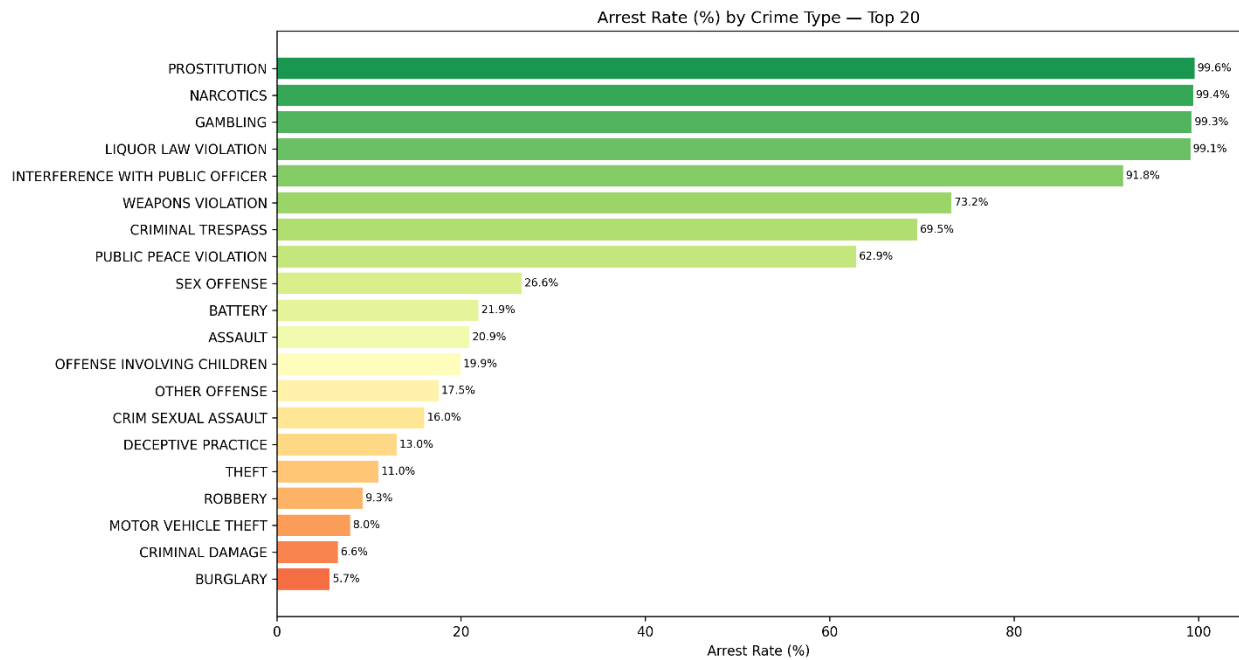


Figure 4: Arrest Rate by Crime Type

2.3.5 Hourly Crime Patterns

Figure 5 illustrates crime occurrence and arrest rates by hour of the day. Crime incidents varied considerably throughout the day. Certain time periods experienced increased criminal activity, reflecting daily human activity patterns and opportunities for crime.

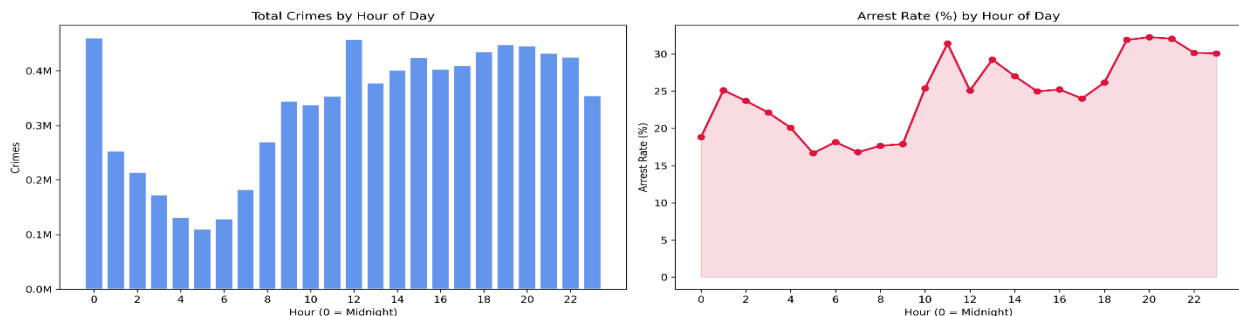


Figure 5: Crimes and Arrest Rate by Hour

2.3.6 Crime by Day of Week

Figure 6 presents crime activity across days of the week. Weekly crime patterns reveal differences between weekdays and weekends. Some days consistently experienced higher crime volumes than others, potentially associated with work schedules, recreational activities, and social behavior patterns.

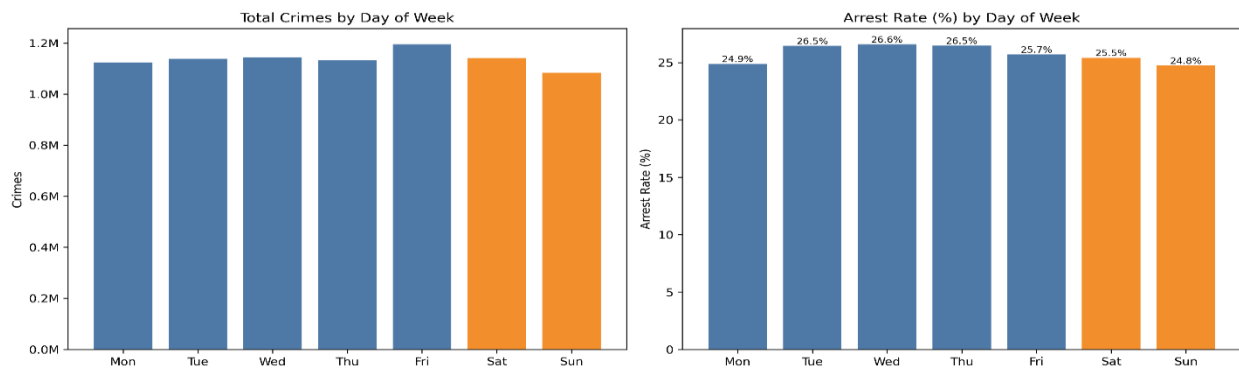


Figure 6: Crimes and Arrest Rate by Day of Week

2.3.7 Spatial Distribution of Crime

Figure 7 displays the top community areas with the highest crime volumes. The spatial analysis indicates that crime is not evenly distributed across Chicago. Certain community areas consistently experience higher crime concentrations than others, supporting the importance of incorporating geographic variables into predictive models.

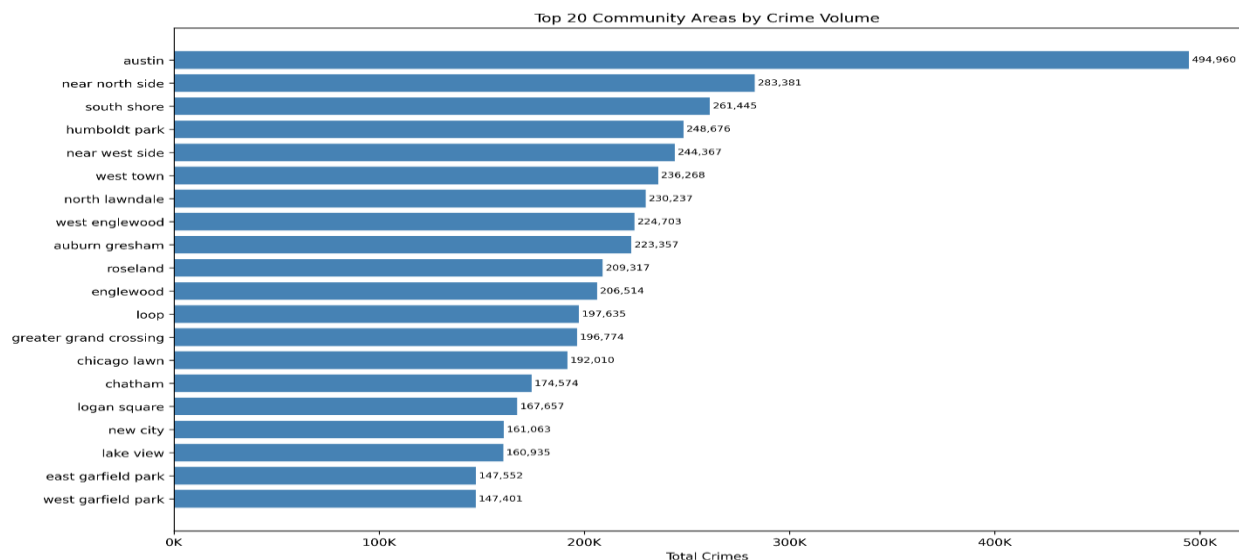


Figure 7: Top Community Areas by Crime Volume

2.3.8 Socio-Economic Factors and Crime

Figure 8 illustrates the relationship between socio-economic indicators and crime levels. The results suggest that socio-economic conditions influence crime patterns across communities. Variables such as unemployment rates, educational attainment, and income levels demonstrate noticeable relationships with crime incidence.

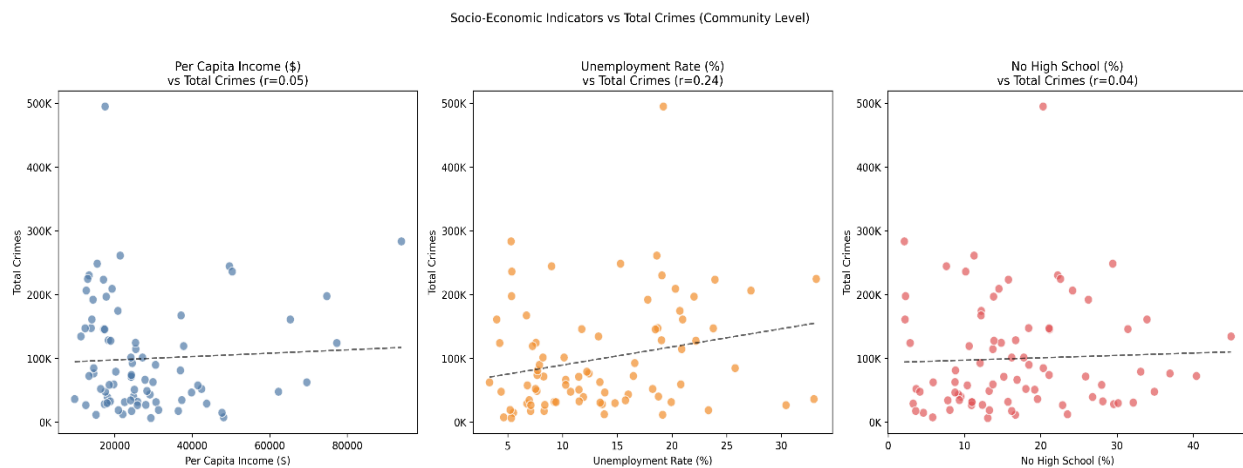


Figure 8: Socio-Economic Factors vs. Total Crimes

2.3.9 Arrest Rates Across Socio-Economic Groups

Figure 9 compares arrest rates across different income and unemployment categories. Variations in arrest outcomes were observed among communities with different socio-economic characteristics, emphasizing the need for community-level analysis.

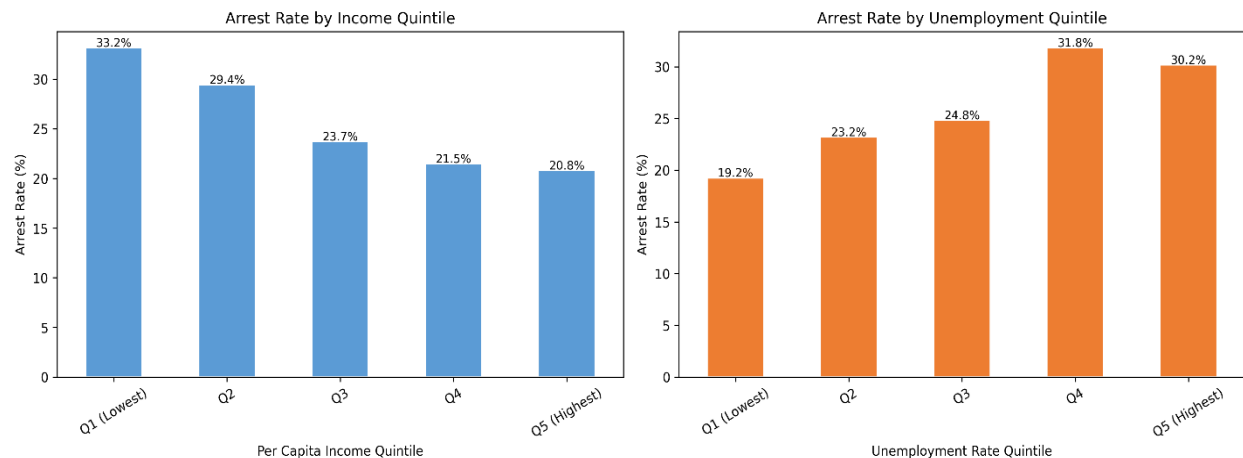


Figure 9: Arrest Rate by Income and Unemployment Quintiles

2.3.10 Weather and Crime Analysis

Figure 10 presents crime counts and arrest rates across temperature categories. Weather conditions appear to influence crime activity. Higher temperatures were generally associated with increased crime levels, while severe weather conditions may reduce outdoor activity and crime opportunities.

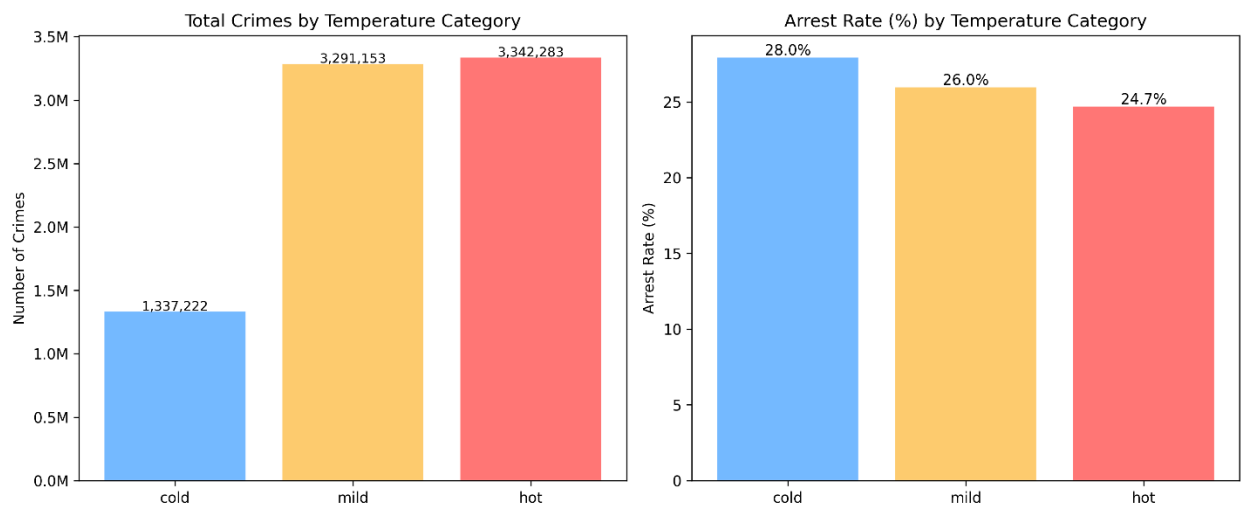


Figure 10: Crimes and Arrest Rate by Temperature Category

2.3.11 Crime and Mobility Analysis

Figure 11 illustrates annual crime counts and public transit ridership trends. The analysis shows a relationship between crime levels and mobility activity. Changes in ridership levels were associated with corresponding changes in crime patterns, suggesting that mobility measures provide valuable contextual information for crime prediction.

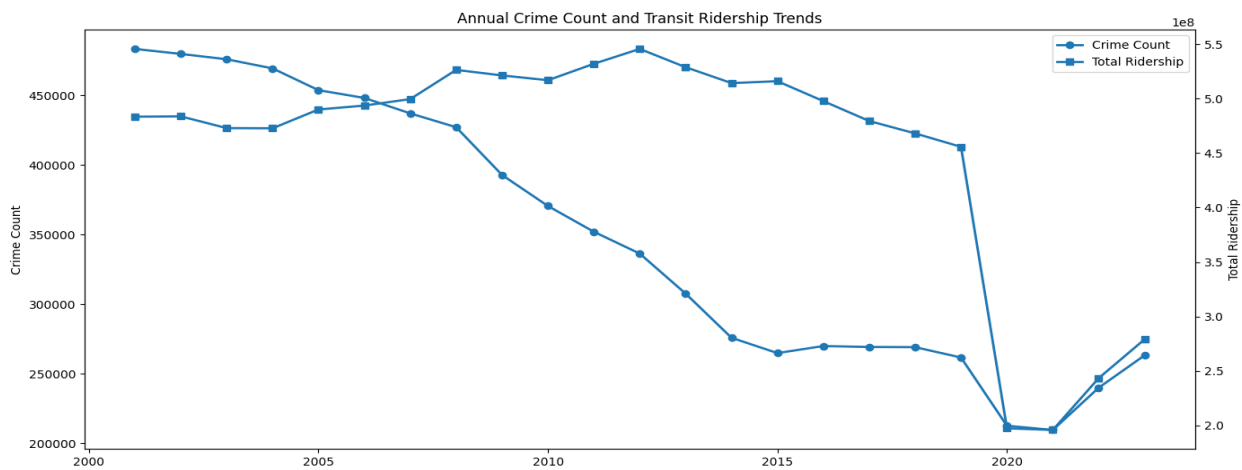


Figure 11: Annual Crime Count and Transit Ridership Trends

2.3.12 Pre-COVID, COVID, and Post-COVID Comparison

Figure 12 compares crime and mobility patterns across three major periods. The COVID-19 pandemic significantly affected both mobility and crime activity. During the pandemic period, public transit ridership declined substantially due to movement restrictions and behavioral changes. Crime patterns also changed during this period, highlighting the influence of societal disruptions on criminal activity.

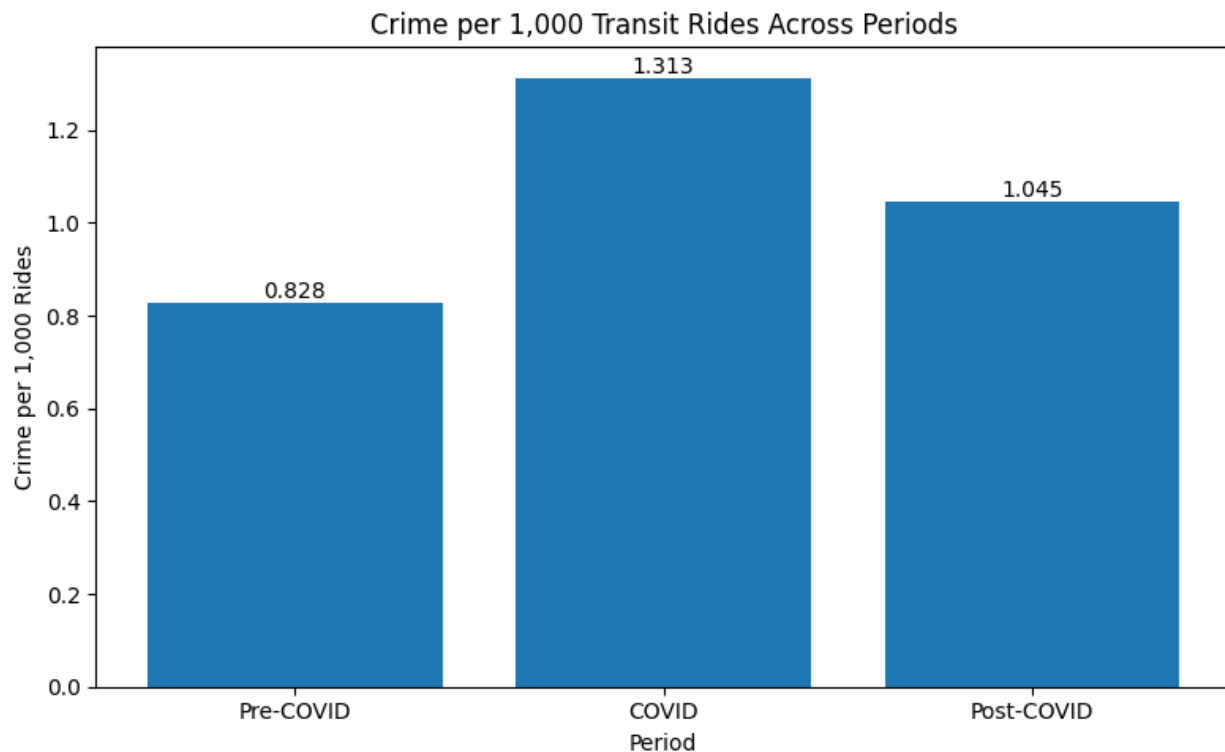


Figure 12: Pre-COVID, COVID, and Post-COVID Comparison

2.3.13 Correlation Analysis

Figure 13 presents the correlation matrix of key variables. The correlation analysis reveals relationships among crime incidents, mobility measures, weather variables, and socio-economic indicators. Several variables demonstrated meaningful associations with arrest outcomes and crime activity, supporting their inclusion in predictive modeling.

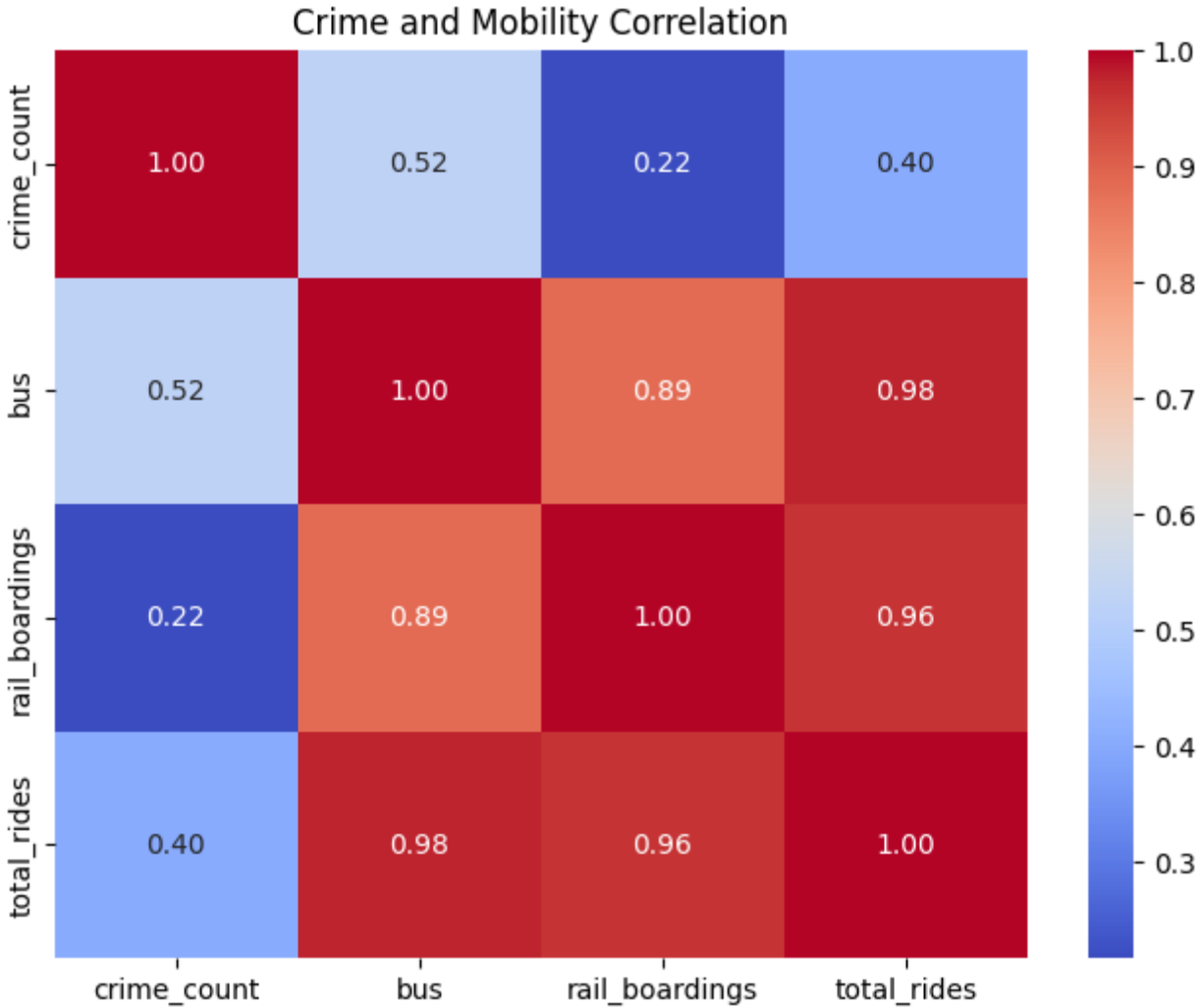


Figure 13: Correlation Heatmap

2.4 Summary of EDA Findings

The exploratory analysis identified important temporal, spatial, socio-economic, environmental, and mobility-related patterns within the dataset. Crime incidents vary across time, location, weather conditions, and levels of human mobility. The findings demonstrate that integrating multiple data sources provides a more comprehensive understanding of crime behavior and arrest outcomes. These observations motivate the use of multi-source machine learning and deep learning models in the subsequent stages of this research.

3. Project Approach

3.1 Research Methodology

This research adopts a data-driven approach for predicting crime arrest outcomes using multiple data sources from the City of Chicago. The methodology integrates crime records, socio-economic indicators, weather information, and public transit mobility data to capture spatial, temporal, environmental, and social factors associated with crime and arrest outcomes.

The overall methodology consists of six major stages: data collection, data preprocessing, exploratory data analysis, feature engineering and selection, model development, and model evaluation.

In the first stage, crime, socio-economic, weather, mobility, and community boundary datasets were collected from publicly available sources. These datasets were then cleaned, transformed, and integrated into a unified analytical framework.

The second stage involved preprocessing operations such as handling missing values, spatial data integration, coordinate-based community area assignment, district imputation, and feature generation. Additional temporal, environmental, and mobility-related variables were created to improve predictive performance.

The third stage consisted of Exploratory Data Analysis (EDA), which was conducted to understand crime trends, arrest patterns, mobility behavior, socio-economic conditions, weather effects, and spatial crime distributions across Chicago communities.

The fourth stage focused on feature engineering and feature selection. Random Forest feature importance and SHAP analysis were used to identify the most influential predictors associated with arrest outcomes.

The fifth stage involved the development of machine learning and deep learning models. Multiple algorithms were trained and evaluated to determine the most effective approach for arrest prediction.

Finally, model performance was assessed using several evaluation metrics, including Accuracy, Precision, Recall, F1-Score, ROC-AUC, and PR-AUC. The best-performing models were then compared and analyzed.

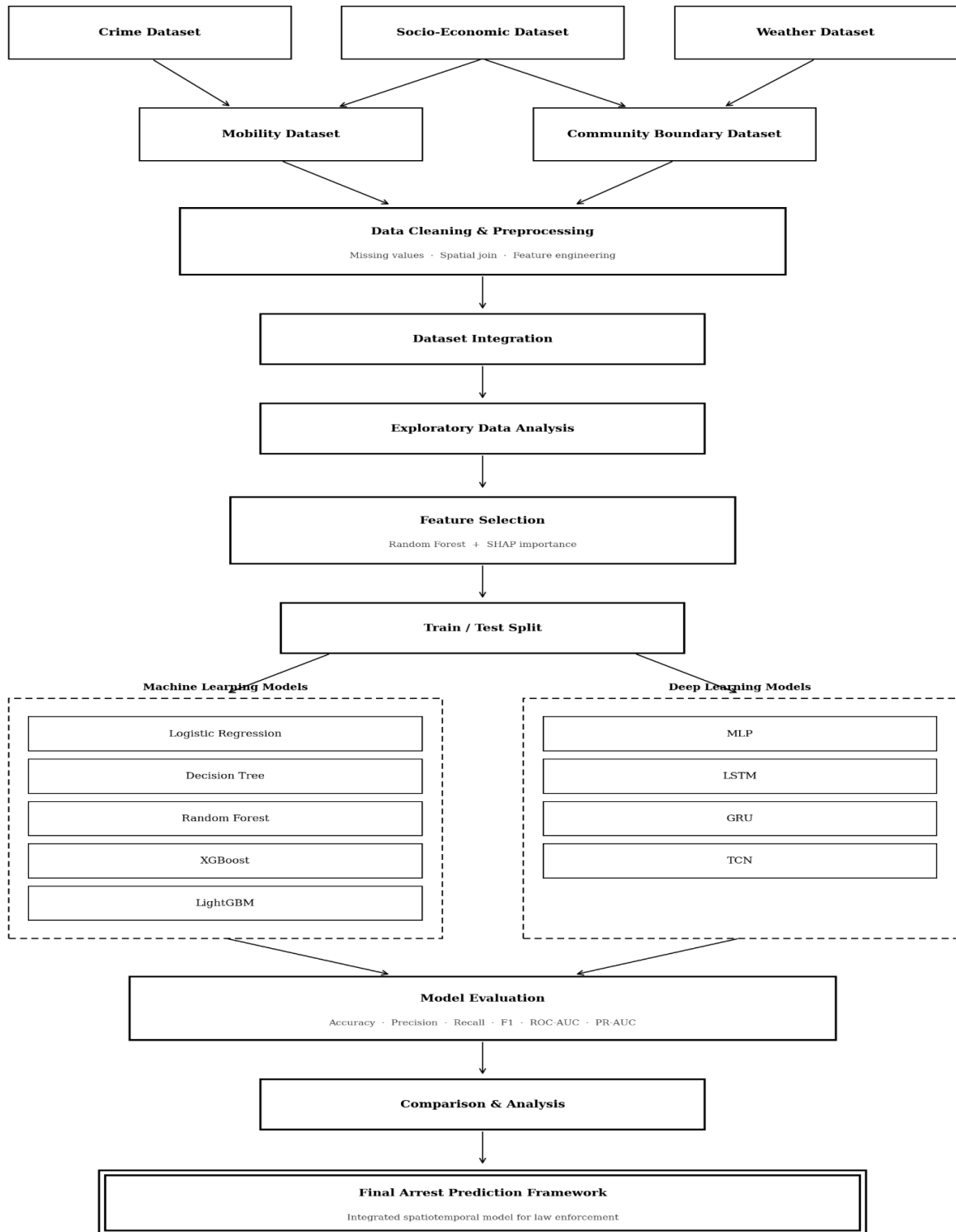


Figure 14: Overall Project Methodology Diagram

3.2 Justification of the Selected Approach

The literature review revealed that most existing studies focus primarily on crime occurrence prediction or hotspot forecasting rather than arrest prediction. While several studies have incorporated weather, socio-economic, or mobility information individually, relatively few studies have integrated all of these factors within a single predictive framework. Furthermore, many existing approaches rely heavily on historical crime records and therefore may not fully capture the broader environmental and social factors that influence crime and arrest outcomes.

Previous studies have demonstrated that spatiotemporal models are effective in capturing complex crime patterns and forecasting future crime occurrences [1], [3], [16]. Similarly, machine learning techniques have shown promising results for arrest prediction and crime analysis [2], [5]. Other researchers have highlighted the importance of mobility data, environmental conditions, weather factors, and socio-economic indicators in explaining crime behavior and improving predictive performance [6], [8], [18]. Additionally, recent advances in deep learning, graph neural networks, and transformer-based architectures have further improved the ability to model spatial and temporal dependencies within crime data [1], [4], [13], [17].

Based on these findings, this research adopts a multi-source data integration strategy that combines crime records, socio-economic indicators, weather information, mobility data, and spatial community characteristics. This approach provides a more comprehensive representation of crime-related conditions than models relying solely on historical crime data. The inclusion of multiple external factors is expected to improve the understanding of arrest outcomes while providing additional contextual information that may influence criminal activity.

Furthermore, previous studies often evaluate only a limited number of algorithms or focus on a specific modeling approach. To address this limitation, this study compares a diverse set of machine learning and deep learning models, including Logistic Regression, Decision Tree, Random Forest, Extra Trees, Gradient Boosting, AdaBoost, XGBoost, LightGBM, CatBoost, Multi-Layer Perceptron (MLP), Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and Temporal Convolutional Network (TCN). Using a common dataset and evaluation framework enables a comprehensive comparison of model performance and provides a more robust assessment of their suitability for arrest prediction.

Finally, this research extends existing literature by investigating arrest outcomes rather than solely focusing on crime occurrence. The study also examines crime and mobility patterns across pre-COVID, COVID, and post-COVID periods, providing additional insights into the relationship between human mobility and crime behavior. Through the integration of multi-source data and advanced predictive modeling techniques, the proposed framework aims to contribute to the development of more accurate and data-driven approaches for crime analytics and public safety decision-making.

3.3 Machine Learning Models

The following machine learning models are used in this study:

- Logistic Regression
- Decision Tree
- Random Forest
- Extra Trees
- Gradient Boosting
- AdaBoost
- XGBoost
- LightGBM
- CatBoost

These models were selected because they are widely used in crime prediction research and provide strong baseline performance for classification tasks.

3.4 Deep Learning Models

The following deep learning models are evaluated:

- Multi-Layer Perceptron (MLP)
- Long Short-Term Memory (LSTM)
- Gated Recurrent Unit (GRU)
- Temporal Convolutional Network (TCN)

These models were selected because previous studies have demonstrated their ability to learn complex temporal and nonlinear relationships from large datasets.

3.5 Model Evaluation

To evaluate model performance, the following metrics are used:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC
- PR-AUC

These metrics provide a comprehensive assessment of classification performance, particularly for imbalanced datasets where accuracy alone may be misleading.

4. GitHub Repository

All project source code, data preprocessing scripts, exploratory data analysis notebooks, feature engineering procedures, machine learning models, deep learning models, and experimental results will be maintained in a GitHub repository.

Repository Link:

https://github.com/Devanshu1013/MRP_Literature_Review

The repository contains:

- Data preprocessing and Crime, socio-economic, weather, and mobility data integration code
- Exploratory Data Analysis (EDA) notebooks
- Feature engineering and feature selection code
- Visualizations and generated figures of EDA
- Project documentation

The repository serves as a centralized platform for code management, reproducibility, and future development of the research project.

References

- [1] Y. Fan, X. Hu, and J. Hu, "Research on a Crime Spatiotemporal Prediction Method Integrating Informer and ST-GCN: A Case Study of Four Crime Types in Chicago," 2025.
- [2] R. Sankul, T. R. Aruva, S. V. Kankal, G. Arrapogula, and S. K. Mohammed, "Crime Data Analysis and Prediction of Arrest Using Machine Learning," 2025.
- [3] X. Han, X. Hu, H. Wu, B. Shen, and J. Wu, "Risk Prediction of Theft Crimes in Urban Communities: An Integrated Model of LSTM and ST-GCN," 2020.
- [4] Riyazuddin, S. Z. Rahman, K. Sreenu, M. Sirajuddin, S. Mazharuddin, and J. Sadiq, "Spatiotemporal Crime Prediction and Forecasting Using Machine Learning and Deep Learning Models," 2026.
- [5] A. Sarker, S. N. Al Hasan, and M. A. Rahman, "Crime Analysis Through Machine Learning," 2021.
- [6] X. Wang, Y. Zhang, and H. Chen, "Exploring Spatio-Temporal and Cross-Type Correlations for Crime Prediction," 2018.
- [7] A. A. Adewoyin and M. A. Khan, "Deep Learning for Real-Time Crime Forecasting and Its Ternarization," 2019.
- [8] S. Kang and H. Kang, "Prediction of Crime Occurrence from Multi-Modal Data Using Deep Learning," PLoS ONE, vol. 12, no. 4, 2017.
- [9] J. Wang, P. Kifer, and D. B. Work, "Predictability Comparison of Three Kinds of Robbery Crime Events Using LSTM," 2019.
- [10] M. Al-Janabi, A. Al-Shourbaji, and M. Shojafar, "Crime Spatiotemporal Prediction with Fused Objective Function in Time Delay Neural Network," 2020.
- [11] J. Zhang, Y. Zheng, and D. Qi, "Deep Spatio-Temporal Residual Networks for Citywide Crowd Flows Prediction," in Proc. AAAI Conf. Artificial Intelligence, 2017.
- [12] M. A. Sarker, M. M. Islam, and S. Islam, "Machine Learning in Crime: A Systematic Literature Review," 2023.
- [13] J. Stec and M. Klabjan, "Examining Deep Learning Architectures for Crime Classification and Prediction," 2018.

- [14] A. M. Ashby, "Impact of COVID-19 on Crime: A Spatial Temporal Analysis in Chicago," 2021.
- [15] L. E. Cohen and M. Felson, "Social Change and Crime Rate Trends: A Routine Activity Approach," *American Sociological Review*, vol. 44, no. 4, pp. 588–608, 1979.
- [16] X. Yu, Z. Chen, and Y. Li, "Crime Hotspot Forecasting: A Recurrent Model with Spatial and Temporal Information," 2017.
- [17] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, and W. Zhang, "Informer: Beyond Efficient Transformer for Long Sequence Time-Series Forecasting," in *Proc. AAAI Conf. Artificial Intelligence*, vol. 35, no. 12, pp. 11106–11115, 2021.
- [18] Y. Wang, C. Liu, Z. Liu, and X. Xie, "Crime Rate Inference with Big Data," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining (KDD)*, pp. 635–644, 2016.